

- Combine FASTA
- EMBL to FASTA
- EMBL Feature Extractor
- EMBL Trans Extractor
- Filter DNA
- Filter Protein
- GenBank to FASTA
- GenBank Feature Extractor
- GenBank Trans Extractor
- One to Three
- Range Extractor DNA
- Range Extractor Protein
- Reverse Complement
- Split Codons
- Split FASTA
- Three to One
- Window Extractor DNA
- Window Extractor Protein

- Codon Plot
- Codon Usage
- CpG Islands
- DNA Molecular Weight

- Fuzzy Search DNA
- Fuzzy Search Protein
- Ident and Sim
- Multi Rev Trans

- ORF Finder
- Pairwise Align Codons
- Pairwise Align DNA
- Pairwise Align Protein

- Protein GRAVY
- Protein Isoelectric Point
- Protein Molecular Weight
- Protein Pattern Find

- Restriction Digest
- Restriction Summary
- Reverse Translate
- Translate

- Color Align Conservation
- Color Align Properties
- Group DNA
- Group Protein
- Primer Map

- Translation Map
- Random Sequences*
- Mutate DNA
- Mutate Protein

- Random DNA Sequences
- Random DNA Regions
- Random Protein Sequences
- Random Protein Regions

Miscellaneous

- Genetic codes
- Browser compatibility
- Mirror this site
- Use this site off-line

The Sequence Manipulation Suite
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Valid XHTML 1.0; Valid CSS

Pattern:	Times found:	Percentage:
g	225	25.74
a	231	26.43
t	226	25.86
c	192	21.97
n	0	0.00
u	0	0.00
r	0	0.00
y	0	0.00
s	0	0.00
w	0	0.00
k	0	0.00
m	0	0.00
b	0	0.00
d	0	0.00
h	0	0.00

Help

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Filters: [Manage Filters](#)

Accession: JF900426.1 GI: 339716221

[Turn Off](#) [Clear](#)

🔍 "Anthoxanthum"[Organism] OR Anthoxanthum[All Fields] (85781) Nucleotide

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```
# Name: Christopher A. Lee
# Date: 12/30/2025
# Prof: Dr. Callura
# Course: Bioinformatics – ENBC 455

# Problem 3
from Bio import Entrez
from Bio import SeqIO

Entrez.email = "clee363@terpmail.umd.edu"
handle = Entrez.efetch(db="nucleotide", id=["JX317645", "JQ011276", "NM_001265803", "HM595636", "JQ712981", "JX460804", "NM_013179", "JX491654", "JF927163", "JQ762396"], rettype="fasta")
records = list(SeqIO.parse(handle, "fasta"))
shortest_seq = min(records, key=lambda x: len(x.seq))
print(shortest_seq.format("fasta"))

# Problem 4
with open('hw2_4.txt', 'r') as file:
    seq = file.readline().strip()
    subseq = file.readline().strip()

    for i, char in enumerate(seq):
        if seq[i: i + len(subseq)] == subseq:
            print(i + 1, "", end="")
```

```
>JX317645.1 Culex quinquefasciatus neuropeptide F mRNA, complete cds
ATGGCATCCACAAGCAGCAGCAGCAGAATCAACAACAACCGCCATGCCGTCAGGTCATCA
GCCTCTTCAGCGTTCACCCAGCGACTGCTAATCGGCCTGCTGGTCTGCACCCTGGTGCTG
GATCTTAGCTGCCTGACCGAGGCCCGGCCCCAGGACGATCCCACCTCCGTCGCCGAAGCC
ATCCGACTGCTGCAGGAGCTGGAAACCAAGCACGCCCAACATGCCCGACCAAGATTCGGA
AAACGTGGCTATCTCCAGCCGGCAAGTTACGGCCAGGACGAACAGGAGCGAAACTATTAC
AGGATGATTGGCAGGATTCAGCGTTTTCAAGATGAACAGAACGCCGTACTCAACTAA
```

```
11 26 33 62 70 138 146 153 205 257 272 279 305 312 386 410 485 492 499 540 558 604 624
660 679 694 701 732 739 746 776 794 818 883 890 918 %
```